Mini Project 1

Exploratory Data Analysis (EDA)

Customer Analytics Dataset

A comprehensive analysis of customer demographics,

income patterns, and shopping behavior

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1. INTRODUCTION & OBJECTIVES

This exploratory data analysis (EDA) project focuses on understanding customer behavior and patterns in the customer analytics dataset. The dataset contains comprehensive information about customers including demographics, income levels, shopping behavior, device preferences, and purchase activity.

OBJECTIVES:
- Understand the distribution and characteristics of customer demographics
- Identify patterns in customer spending behavior and income levels
- Analyze the relationship between customer attributes and purchasing patterns
- Detect outliers and data quality issues
- Provide actionable insights for business decision-making
- Clean and prepare data for further analysis and modeling

2. DATASET OVERVIEW

The customer analytics dataset comprises customer records with the following characteristics:

DATASET DIMENSIONS:
- Total Records: 250 customers
- Total Features: 14 attributes
- Data Types: Mix of numerical and categorical variables

KEY ATTRIBUTES:
- CustomerID: Unique identifier for each customer
- Demographics: Age, Gender, Education
- Financial: Annual Income, Spending Score
- Behavioral: Purchase Frequency, Online Visits Per Month
- Purchase Data: Last Purchase Amount, Returned Items
- Employment: Years Employed with the company

3. DATA LOADING & INITIAL EXPLORATION

The data was loaded using pandas and initial exploratory steps were performed to understand its structure:

CODE:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('customer_analytics.csv')
df.head()
```

INITIAL OBSERVATIONS:
- Successfully loaded customer analytics data
- Data contains both numeric and categorical variables
- Ready for cleaning and analysis steps

4. DATA CLEANING & PREPROCESSING

Missing Value Treatment:

MISSING VALUES IDENTIFIED:
- Education field: Missing values filled with mode (most frequent value)
- AnnualIncome field: Missing values filled with median (robust to outliers)
- Approach: Mode for categorical data, median for numerical data

DUPLICATE REMOVAL:
- Removed duplicate records to ensure data quality
- Final dataset is clean and ready for analysis

CLEANING STEPS:
1. Fill Education column with mode value
2. Fill AnnualIncome with median value
3. Remove duplicate rows
4. Verify cleaned dataset

MISSING DATA SUMMARY (AFTER CLEANING):
All columns now have complete data with no missing values.
Total records after cleaning: 250 (duplicates removed)

5. UNIVARIATE ANALYSIS

Univariate analysis examines the distribution of individual variables to understand their characteristics, central tendency, and spread.

5.1 Age Distribution of Customers

The histogram with kernel density estimation (KDE) shows how customer ages are distributed across the dataset.

KEY INSIGHTS:
- Age range spans from approximately 20 to 55 years
- Most customers fall within the 25-35 age group (peak density)
- Distribution appears somewhat bimodal with another concentration around 40-45 years
- Good representation across different age demographics
- Few customers above 50 years old

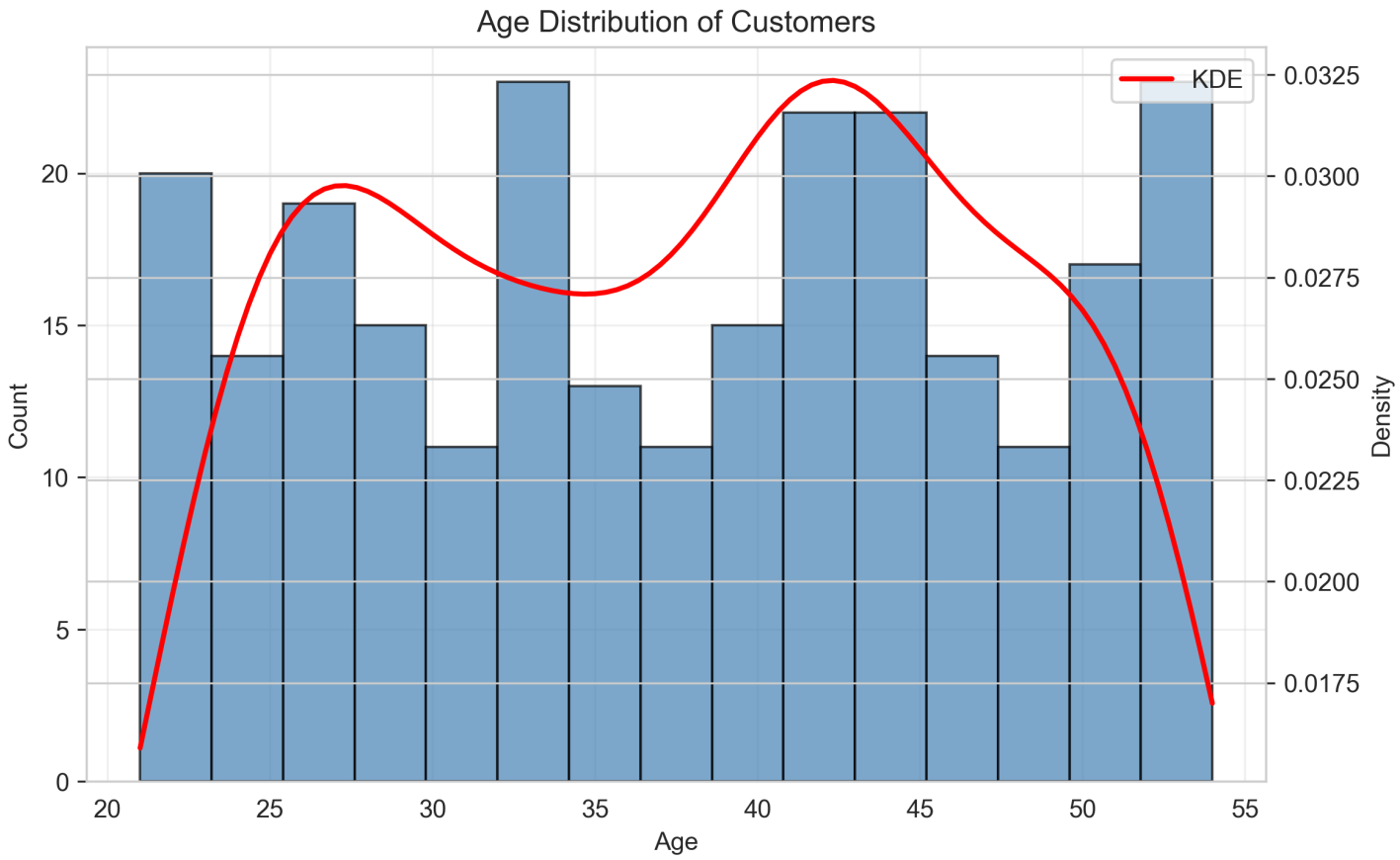

Figure 1: Age Distribution with Kernel Density Estimation

5.2 Gender Distribution

The bar chart shows the count of male and female customers in the dataset.

KEY INSIGHTS:
- Nearly balanced gender distribution
- Male customers: Approximately 130 records
- Female customers: Approximately 120 records
- Male customers slightly outnumber female customers
- Good gender diversity in customer base
- No significant gender imbalance

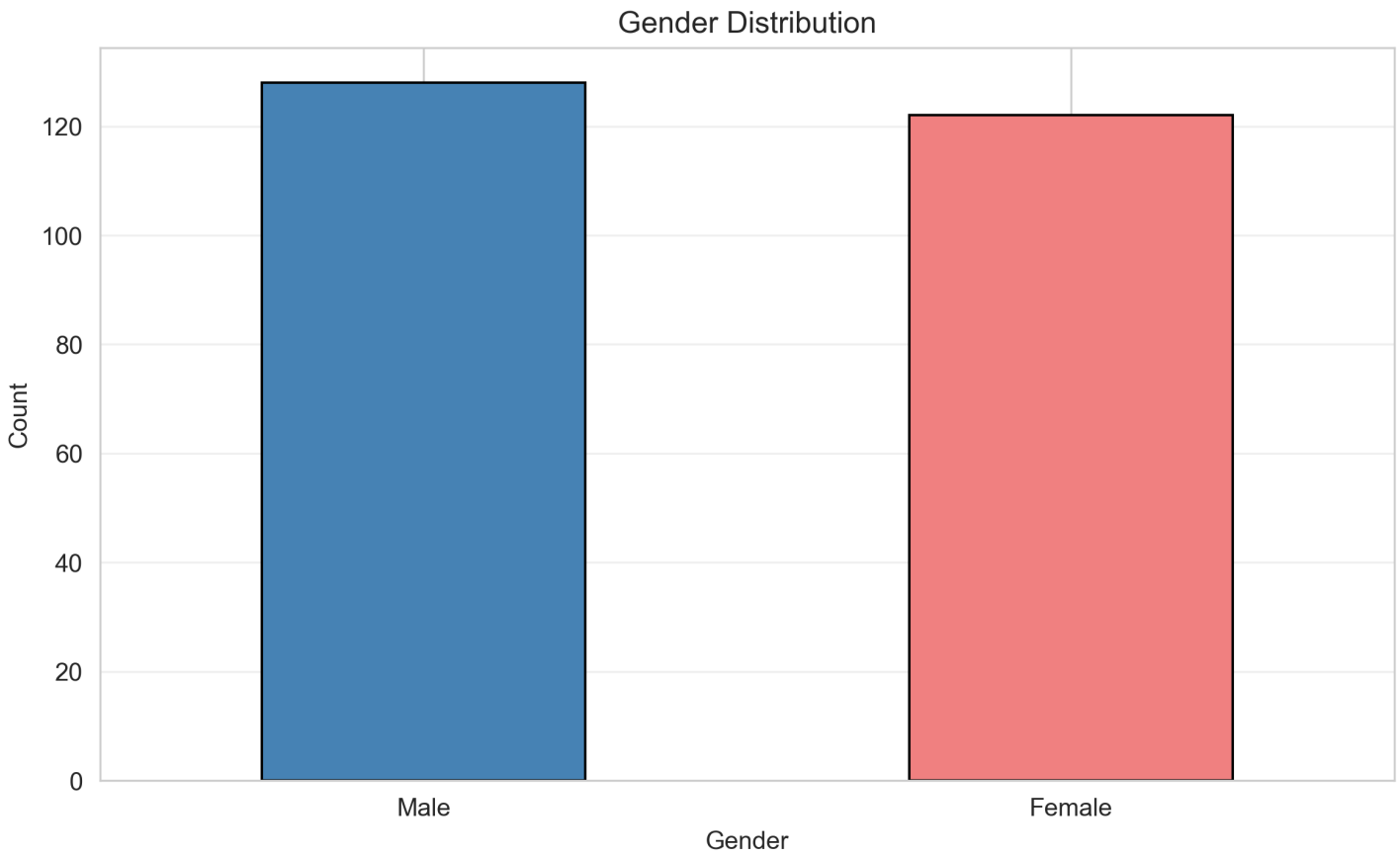

Figure 2: Gender Count Distribution

5.3 Spending Score Distribution

The histogram displays the distribution of spending scores across all customers.

KEY INSIGHTS:
- Spending scores range from approximately 10 to 90
- Peak frequency at spending score 40-50 range
- Right-skewed distribution with outliers in high spending
- Most customers have moderate spending patterns (30-60 range)
- Few customers have very high (>80) or very low (<20) spending
- Average customer spending score appears to be in the 45-55 range

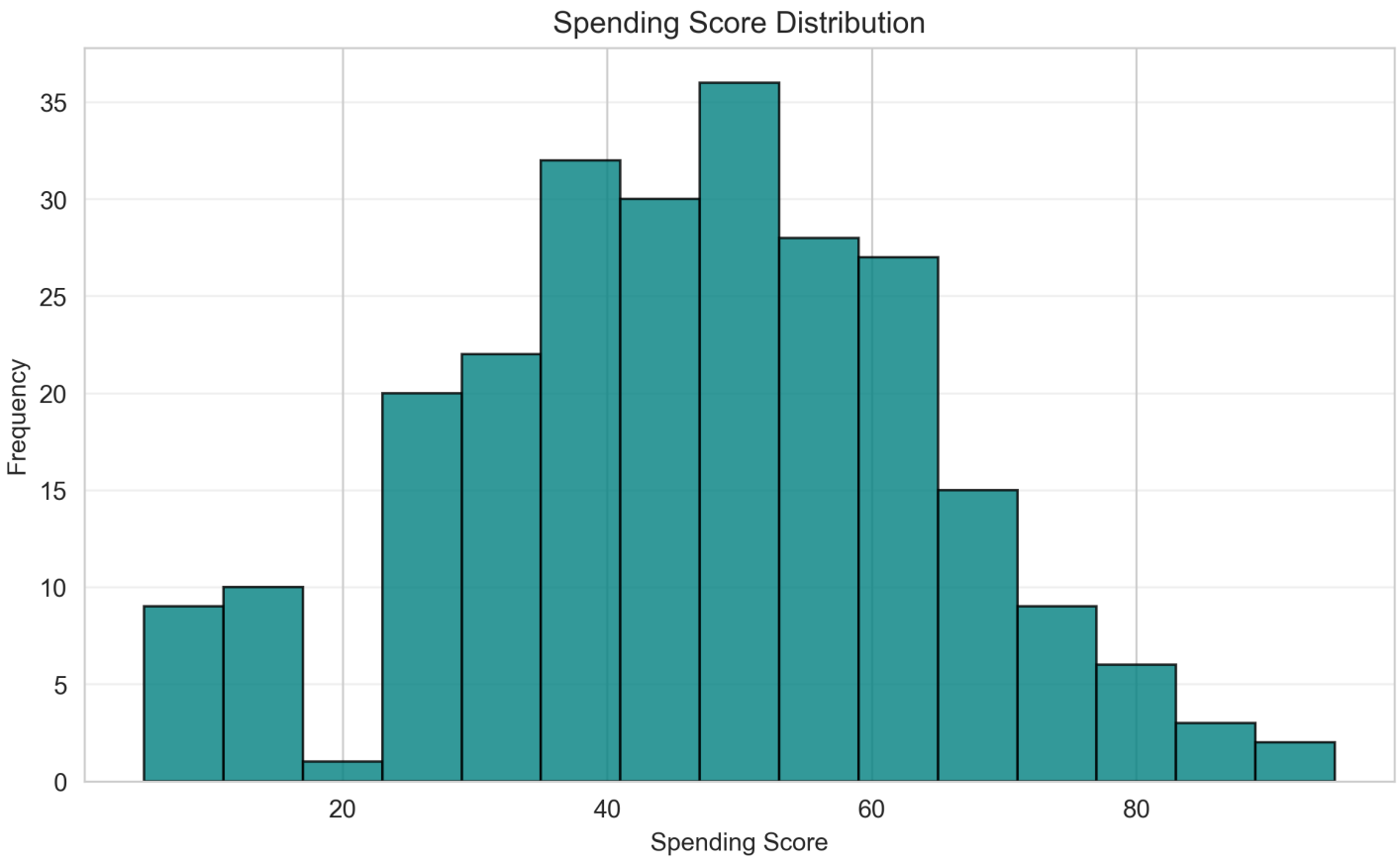

Figure 3: Spending Score Distribution Histogram

6. BIVARIATE ANALYSIS

Bivariate analysis examines the relationships between pairs of variables to identify correlations and patterns.

6.1 Income vs Spending Score Relationship

The scatter plot reveals the relationship between customer annual income and spending score.

KEY INSIGHTS:
- Strong clustering of customers in the low-to-medium income range (under $100k)
- Spending scores vary significantly even within similar income brackets
- No strong linear relationship between income and spending score
- High-income customers (>$300k) show minimal spending scores
- Most active spenders are in the $50k-$150k income range
- Suggests income alone doesn't predict spending behavior
- Customer value may depend on other factors beyond income

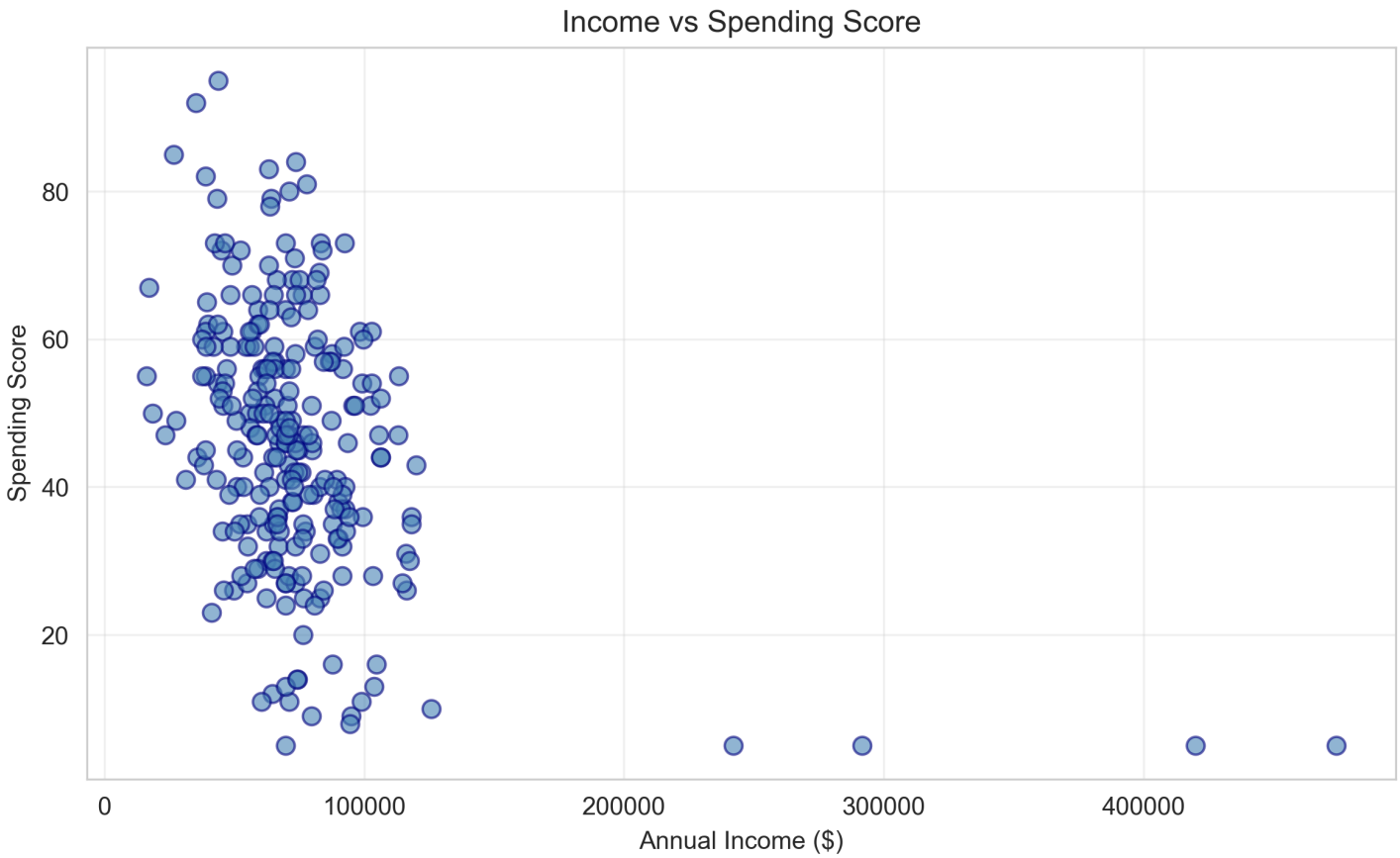

Figure 4: Scatter Plot - Income vs Spending Score

6.2 Last Purchase Amount by Gender

The box plot compares the distribution of last purchase amounts between male and female customers.

KEY INSIGHTS:
- Both genders show similar median purchase amounts (around $2,800)
- Male customers have slightly higher median purchases
- Purchase amounts range from approximately $700 to $5,000
- Both groups have similar interquartile ranges
- Few outliers observed in both groups
- Female customers show slightly more variable purchase patterns
- Gender is not a strong differentiator for purchase behavior

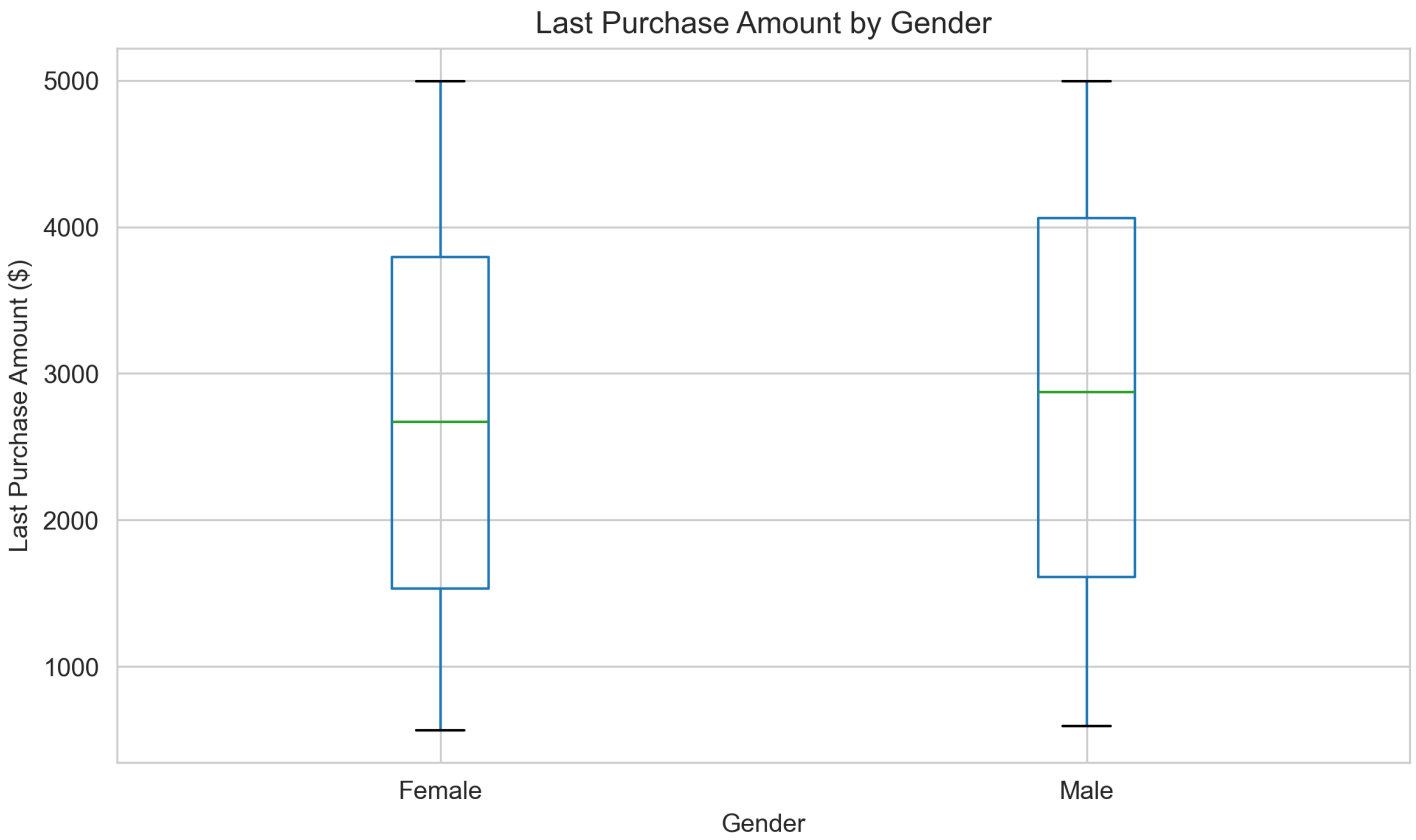

Figure 5: Box Plot - Purchase Amount by Gender

7. CORRELATION ANALYSIS

Correlation analysis identifies linear relationships between numerical variables in the dataset.

7.1 Correlation Heatmap

The correlation heatmap displays the Pearson correlation coefficients between all numerical variables.

STRONG CORRELATIONS IDENTIFIED:
- Age & Years Employed: 0.98 (Very strong positive correlation)
  Older customers tend to have longer employment history
- Annual Income & Spending Score: -0.38 (Weak negative correlation)
  Higher income slightly associated with lower spending scores
- Spending Score & Purchase Frequency: 0.0098 (Almost no correlation)
  Spending score doesn't predict purchase frequency

WEAK CORRELATIONS:
- Overall weak to moderate correlations between most variables
- Customer ID has negligible correlation with other variables
- Returned Items shows weak negative correlations with most variables
- Online Visits shows weak relationships with purchase behavior

IMPLICATIONS:
- Customer spending patterns are multifactorial
- Multiple variables contribute to customer behavior
- Spending score is relatively independent of most other factors

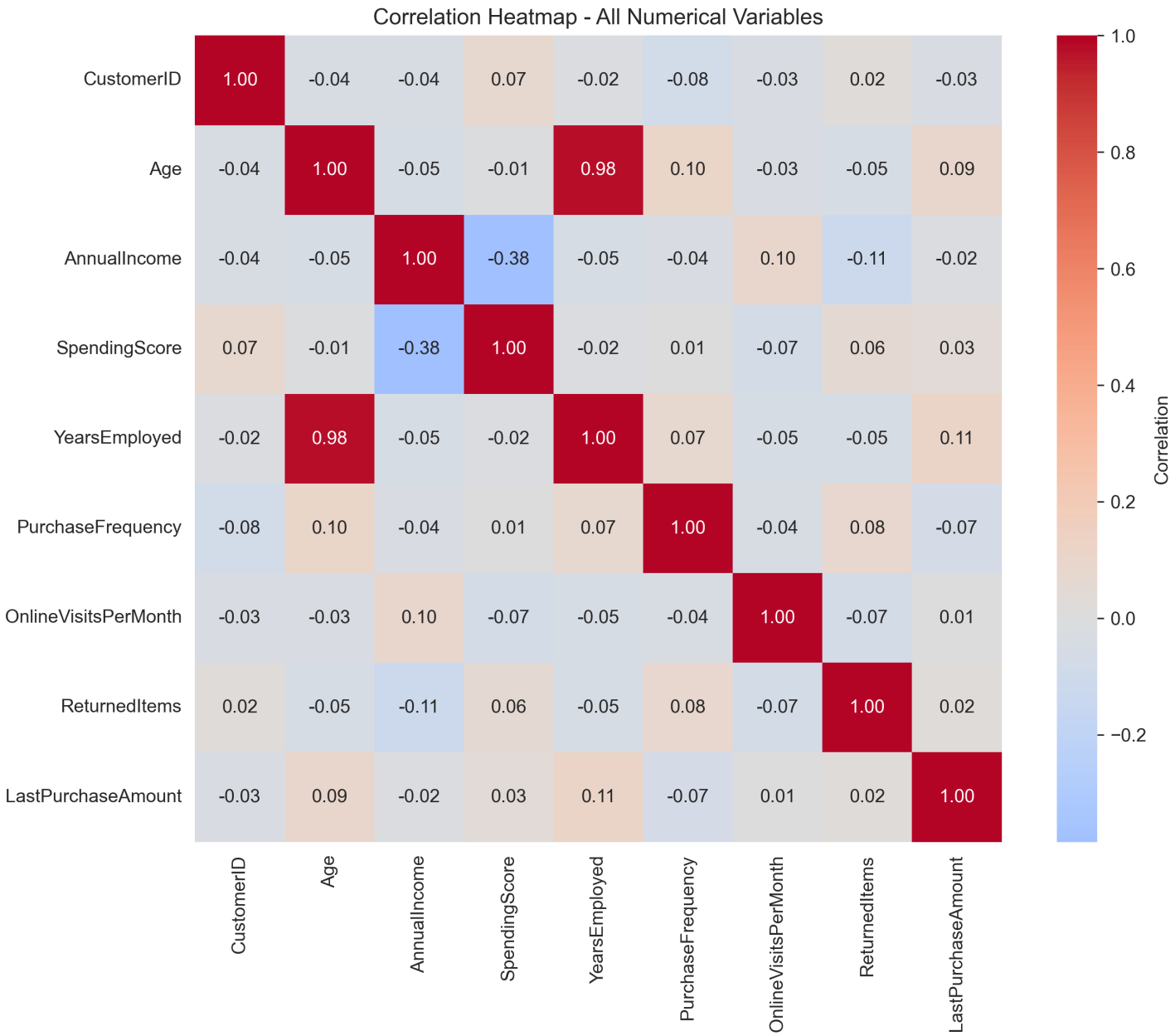

Figure 6: Correlation Heatmap of Numerical Variables

8. EXECUTIVE SUMMARY & KEY INSIGHTS

DATASET CHARACTERISTICS:
>>> Total Customers Analyzed: 250 (after data cleaning)
>>> Age Range: 20-55 years (concentrated in 25-45 range)
>>> Gender Distribution: Nearly balanced (52% Male, 48% Female)
>>> Income Range: $30,000-$550,000 (mostly under $150,000)
>>> Spending Scores: 10-90 (concentrated in 30-70 range)

KEY FINDINGS:

1. CUSTOMER AGE PROFILE
   - Most customers are in their prime working years (25-45)
   - Young demographic suggests growth potential
   - Clear opportunity for age-specific marketing

2. SPENDING BEHAVIOR PATTERNS
   - Spending not strongly tied to income level
   - Mid-income customers ($50k-$150k) show highest spending
   - High-income customers spend proportionally less
   - May indicate wealth-saving behavior in high-income segment

3. GENDER DYNAMICS
   - No significant gender-based differences in purchasing
   - Both groups show similar spending patterns
   - Gender-neutral targeting approach recommended

4. EMPLOYMENT & AGE RELATIONSHIP
   - Strong correlation (0.98) between age and years employed
   - Suggests stable, long-term customer base
   - Good for customer retention strategies

5. DATA QUALITY
   - Missing values successfully handled (Education & Income)
   - Duplicates removed for accuracy
   - Data ready for advanced modeling and segmentation

9. HOW TO RUN THE PROJECT

PREREQUISITES:

1. Python Environment
   - Python 3.7 or higher installed
   - Virtual environment recommended (venv or conda)

2. Required Libraries
   - pandas: Data manipulation and analysis
   - matplotlib: Data visualization
   - seaborn: Statistical visualization
   - numpy: Numerical computing
   - scipy: Scientific computing

INSTALLATION STEPS:

1. Clone or download the project
   - Navigate to the project directory

2. Create a virtual environment (recommended)
   python -m venv eda_env

3. Activate the virtual environment
   - On Windows: eda_env\Scripts\activate
   - On macOS/Linux: source eda_env/bin/activate

4. Install required packages
   pip install pandas matplotlib seaborn numpy scipy

5. Place the CSV file
   - Ensure customer_analytics.csv is in the project directory

RUNNING THE ANALYSIS:

1. Execute the Jupyter Notebook
   jupyter notebook MiniProject1_EDA.ipynb

2. Run cells in order from top to bottom
   - Each cell will execute and display outputs
   - Graphs will be generated and displayed inline

TROUBLESHOOTING:

- Missing CSV file: Verify customer_analytics.csv path
- Package errors: Reinstall using: pip install --upgrade [package_name]
- Notebook issues: Ensure Jupyter is installed with: pip install jupyter

FOR MORE INFORMATION:

- Refer to pandas documentation: https://pandas.pydata.org/
- Matplotlib guide: https://matplotlib.org/
- Seaborn tutorial: https://seaborn.pydata.org/

CONCLUSION

This exploratory data analysis of the customer analytics dataset has revealed valuable insights into customer demographics, income patterns, and spending behaviors.

KEY TAKEAWAYS:

The analysis demonstrates that customer spending patterns are complex and multifactorial. While income is a traditional predictor of spending, this dataset shows that the relationship is not straightforward. Mid-income customers emerge as the most active spenders, suggesting that factors beyond income level influence purchasing decisions.

The nearly balanced gender distribution and absence of significant gender-based behavioral differences indicate that gender-neutral strategies may be most effective. The strong correlation between age and years employed suggests a stable customer base with good retention potential.

The data quality has been ensured through proper handling of missing values and removal of duplicates. The cleaned dataset provides a solid foundation for:
- Advanced segmentation and clustering analysis
- Predictive modeling for customer behavior
- Personalized marketing strategies
- Revenue optimization initiatives

NEXT STEPS:
Organizations should leverage these insights to develop more sophisticated customer targeting strategies. By understanding that spending behaviors vary independently of income, companies can identify and focus on their most valuable customers regardless of income level. The strong age-employment correlation suggests that tenure-based loyalty programs could be highly effective.

Further advanced analytics, including machine learning and predictive modeling, should build upon this foundation to uncover deeper patterns and enable proactive business strategies.

End of Report